Mock 03

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1 Instructions

- Time Allowed: 120 Minutes.
- Total Marks: 41 (Scaled internally).
- Section A: 10 MCQ (1 point each).
- Section B: 10 Short Open (1.5 points each).
- Section C: 8 Complex Open (2 points each).
- Note: Includes Julia code interpretation and writing.

2 Section A: Multiple Choice (1 Point Each)

1. Which set is countable?
   (a) Functions f: N -> {0,1}
   (b) Infinite bit strings
   (c) Power set of N
   (d) Finite subsets of N

2. Julia Code Output:

```
S = Set([1, 2, 3])
res = Set()
for x in S
    if x % 2 != 0 push!(res, x^2) end
end
```

   (a) Set([1, 4, 9])
   (b) Set([1, 9])
   (c) Set([2])
   (d) Set([4])

3. Negation of P -> Q is:
   (a) P & !Q
   (b) !P | Q
   (c) !P & Q
   (d) P | !Q

4. Locus |z - 1| + |z + 1| = 4 is:
   (a) Circle
   (b) Ellipse
   (c) Segment
   (d) Hyperbola

5. Antisymmetric but NOT reflexive relation on {a,b,c}:
   (a) {(a,a), (b,b), (c,c)}

(b) {(a,b), (b,a)}
(c) {(a,b), (b,c), (a,c)}
(d) {(a,b), (b,a), (a,a)}

6. If a = b (mod m), which is FALSE?

   (a) a+c = b+c
   (b) ac = bc
   (c) $a^2 = b^2$
   (d) $c^a = c^b$

7. NOT a vector space:

   (a) Polynomials deg=3
   (b) Polynomials deg<=3
   (c) Continuous functions
   (d) Solutions to y"+y=0

8. Integer solutions x1+x2+x3=10, xi>=1:

   (a) C(12,2)
   (b) C(9,2)
   (c) C(10,3)
   (d) 3^{10}

9. Bounded sequence implies:

   (a) Convergence
   (b) Convergent subsequence
   (c) Monotonicity
   (d) Finite range

10. Units in Z_{10}:

    (a) {1,3,7,9}
    (b) {1,2,3,4,5}
    (c) All elements
    (d) {1,5}

3 Section B: Short Open Problems (1.5 Points Each)

3.1 Problem B1

Write a Julia function ‘CountDivisible(n, k)‘ that counts numbers in 1:n divisible by k but NOT by k^2.

3.2 Problem B2

Simplify $(P \to Q) \wedge (P \to \neg Q)$ to show it implies $\neg P$.

3.3 Problem B3

Calculate $(1+i)^{10}$ in the form $a + bi$.

3.4 Problem B4

Prove or disprove: $A \setminus (B \cup C) = (A \setminus B) \cup (A \setminus C)$.

3.5 Problem B5

Permutations of “SUCCESS” with ’C’s NOT adjacent.

3.6 Problem B6

Find $3^{100} \pmod 7$.

3.7 Problem B7

Remainder of $P(x) = x^{2025} - x + 1$ divided by $x - i$.

3.8 Problem B8

Prove $f(x) = x^3 + x$ is injective.

3.9 Problem B9

Projection of $(3, 1, 2)$ onto $(1, -1, 0)$.

3.10 Problem B10

Limit of $\sqrt{n^2 + 5n} - n$.

4 Section C: Complex Open Problems (2 Points Each)

4.1 Problem C1

Prove $\sum_{i=1}^{n} i \cdot i! = (n+1)! - 1$.

4.2 Problem C2

Prove that if $a^2 = a$ for all a in ring R, then R is commutative.

4.3 Problem C3

Prove the set of circles in plane with rational center and radius is countable.

4.4 Problem C4

Prove $\lim \frac{3n^2+1}{2n^2-n} = 1.5$ using $\epsilon - N$.

4.5 Problem C5

Solve system: $2x \equiv 1 \pmod 5$ AND $3x \equiv 2 \pmod 7$.

4.6 Problem C6

If $|z_k| = 1$, prove $\arg(z_1/z_2) = 2\arg((z_1 - z_3)/(z_2 - z_3))$.

4.7 Problem C7

Prove f injective iff $f(A \cap B) = f(A) \cap f(B)$ for all subsets.

4.8 Problem C8

Distance between skew lines: r = (1,0,0) + t(1,1,0) r = (0,1,1) + s(0,1,1)

5 Section D: Answer Key & Grading Rubric

5.1 Part A: MCQ Answers

Q	Ans	Topic
1	D	Cardinality
2	B	Set Building
3	A	Logic (Negation)
4	B	Conic Sections (Ellipse)
5	C	Relations
6	D	Modular Arithmetic
7	A	Vector Spaces (Polynomials)
8	B	Combinatorics
9	B	Analysis (Bolzano-Weierstrass)
10	A	Number Theory (Coprimality)

5.1.1 Explanatory Notes

- **Q1:** Finite subsets are countable. The set of functions/infinite strings maps to $2^{\aleph_0}$ (uncountable).
- **Q2:** Input $\{1, 2, 3\}$. 1 is odd $\to 1^2 = 1$; 2 is even $\to$ skip; 3 is odd $\to 3^2 = 9$. Result $\{1, 9\}$.
- **Q4:** Definition of an ellipse: sum of distances to foci is constant ($2a$). Here, distance is $4 > 2$ (dist between foci), so it is a valid ellipse.
- **Q7:** Degree **exactly** n is not a subspace because it fails closure under addition: $(x^3 + x) + (-x^3 + 1) = x + 1$ (degree 1).

5.2 Part B: Short Solutions & Rubric

5.2.1 B1: Julia Function

```
function count_divisible(n, k)
    count(x -> (x % k == 0) && (x % k^2 != 0), 1:n)
end
```

- ☐ **(0.5 pts)** Function definition syntax.
- ☐ **(0.5 pts)** Correct modulo logic (‘% k = 0‘ AND ‘% k^2 ! 0‘).
- ☐ **(0.5 pts)** Correct usage of range and count/loop.

5.2.2 B2: Logic Simplification

Goal: Simplify $(P \to Q) \wedge (P \to \neg Q)$.

1. Equivalence: $(\neg P \vee Q) \wedge (\neg P \vee \neg Q)$.
2. Distributive Law: $\neg P \vee (Q \wedge \neg Q)$.
3. Complement: $\neg P \vee F \equiv \neg P$.

Conclusion: This is false only if P is true.

- ☐ **(1.5 pts)** 0.5 points per logical step/justification.

5.2.3 B3: Complex Power

$1 + i = \sqrt{2}e^{i\pi/4}$. $(1 + i)^{10} = (\sqrt{2})^{10}e^{i10\pi/4} = 32e^{i5\pi/2}$. Reduce angle: $5\pi/2 \equiv \pi/2 \pmod{2\pi}$. Result: $32e^{i\pi/2} = 32i$.

- ☐ **(0.5 pts)** Polar form conversion.
- ☐ **(0.5 pts)** Exponent calculation.
- ☐ **(0.5 pts)** Final rectangular form.

5.2.4 B4: Set Theory Disproof

Statement: $(A \setminus B) \cup C = (A \cup C) \setminus (B \cup C)$. **False. Counterexample:** Let $A = \{1\}, B = \{1\}, C = \emptyset$.

- LHS: $1 \setminus \{1\} = \emptyset$.
- RHS: $\emptyset \cup \{1\} = \{1\}$.

Since $\emptyset \neq \{1\}$, the statement is false.

- ☐ **(0.5 pts)** Stating False.
- ☐ **(1.0 pt)** Valid counterexample.

5.2.5 B5: Combinatorics (Mississippi Style)

Total permutations of letters (C, C, S, S, S, ... assume 7 total, 2 C, 3 S): $\frac{7!}{2!3!}$. **Restriction:** CC must not be adjacent. Total with CC adjacent (treat CC as one unit): $\frac{6!}{3!}$. Result: $\frac{7!}{2!3!} - \frac{6!}{3!}$.

- ☐ **(0.5 pts)** Total arrangements.
- ☐ **(0.5 pts)** Adjacent (bundled) arrangements.
- ☐ **(0.5 pts)** Subtraction principle.

5.2.6 B6: Fermat's Little Theorem

Find $3^{100} \pmod 7$. By FLT, $3^6 \equiv 1 \pmod 7$. $100 = 6(16) + 4$. $3^{100} \equiv (3^6)^{16} \cdot 3^4 \equiv 1^{16} \cdot 81 \pmod 7$. $81 = 7(11) + 4 \implies 81 \equiv 4 \pmod 7$.

- ☐ **(0.5 pts)** Exponent breakdown using FLT.
- ☐ **(0.5 pts)** Calculation of remainder.
- ☐ **(0.5 pts)** Final answer (4).

5.2.7 B7: Polynomial Remainder

Evaluate $P(i) = i^{2025} - i + 1$. Powers of i: $2025 = 4(506) + 1$, so $i^{2025} = i^1 = i$. $P(i) = i - i + 1 = 1$.

- ☐ **(0.5 pts)** Usage of Remainder Theorem logic.
- ☐ **(0.5 pts)** Reduction of i power.
- ☐ **(0.5 pts)** Final result.

5.2.8 B8: Injectivity Proof

Assume $f(x) = f(y)$. $x^3 + x = y^3 + y \implies (x^3 - y^3) + (x - y) = 0$. $(x - y)(x^2 + xy + y^2 + 1) = 0$. Check factor: $x^2 + xy + y^2 + 1 = (x + y/2)^2 + \frac{3y^2}{4} + 1$. Since squares are non-negative, the term is $\geq 1 > 0$. Therefore, $x - y$ must be 0, so $x = y$.

- ☐ **(0.5 pts)** Setup $f(x) = f(y)$.
- ☐ **(0.5 pts)** Factorization.
- ☐ **(0.5 pts)** Showing quadratic factor is strictly positive.

5.2.9 B9: Vector Projection

$\text{proj}_{\mathbf{u}}\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{u}}{|\mathbf{u}|^2}\mathbf{u}$. $\mathbf{v} = (1, 2, 0), \mathbf{u} = (1, -1, 0)$. $\mathbf{v} \cdot \mathbf{u} = 1 - 2 + 0 = -1$. (Wait, key says $3 - 1 = 2$? Assuming input typo, strictly following key math: $\mathbf{v} \cdot \mathbf{u} = 2$, $|\mathbf{u}|^2 = 2$). Result: $\frac{2}{2}(1, -1, 0) = (1, -1, 0)$.

- ☐ **(0.5 pts)** Formula.
- ☐ **(0.5 pts)** Dot product/Norm calc.
- ☐ **(0.5 pts)** Final vector.

5.2.10 B10: Limits of Radicals

Limit: $\lim_{n\to\infty}(\sqrt{n^2+5n}-n)$. Multiply by conjugate $\frac{\sqrt{n^2+5n}+n}{\sqrt{n^2+5n}+n}$. Numerator: $(n^2+5n)-n^2=5n$. Denominator: $n\sqrt{1+5/n}+n=n(\sqrt{1+5/n}+1)$. Limit: $\frac{5n}{n(1+1)}=\frac{5}{2}$.

- ☐ **(0.5 pts)** Conjugate step.
- ☐ **(0.5 pts)** Algebra simplification.
- ☐ **(0.5 pts)** Final limit.

5.3 Part C: Complex Solutions (Sketch)

5.3.1 C1: Factorial Inequality Induction

Goal: Show $\sum_{i=1}^{n} i(i!)=(n+1)!-1$.

1. **Base Case ($n=1$):** $1(1!)=1$. RHS: $2!-1=1$. Valid.
2. **Step:** Assume $\sum_{i=1}^{k} i(i!)=(k+1)!-1$. Add $(k+1)(k+1)!$ to both sides. RHS becomes: $((k+1)!-1)+(k+1)(k+1)!$ Factor out $(k+1)!$: $(k+1)![1+(k+1)]-1=(k+1)!(k+2)-1=(k+2)!-1$.
3. **Conclusion:** True for all n.

5.3.2 C2: Boolean Rings

Given ring where $a^2=a$ for all a (Boolean Ring).

1. Consider $(a+b)^2=a+b$. $a^2+ab+ba+b^2=a+b$. $a+ab+ba+b=a+b \implies ab+ba=0 \implies ab=-ba$.
2. Consider $2a$. Since $(2a)^2=2a$: $4a^2=2a \implies 4a=2a \implies 2a=0$. Thus $a=-a$.
3. Substitute into (1): $ab=-ba=ba$. The ring is commutative.

5.3.3 C3: Countability of Circles

A circle is determined by center (x,y) and radius r. The set of parameters is $(x,y,r)\in\mathbb{Q}^3$. Since $\mathbb{Q}$ is countable, the Cartesian product $\mathbb{Q}\times\mathbb{Q}\times\mathbb{Q}$ is countable. Therefore, the set of circles with rational centers and radii is countable.

5.3.4 C4: Epsilon-Delta Limit Proof

Limit to prove: $\lim_{n\to\infty} \frac{3n^2+1}{2n^2-n} = \frac{3}{2}$.

$$\left|\frac{3n^2+1}{2n^2-n} - \frac{3}{2}\right| = \left|\frac{2(3n^2+1) - 3(2n^2-n)}{2(2n^2-n)}\right|$$

$$= \left|\frac{6n^2+2-6n^2+3n}{4n^2-2n}\right| = \left|\frac{3n+2}{4n^2-2n}\right|$$

Bounding: For large n, $3n+2 < 4n$ and $4n^2 - 2n > 2n^2$ (implies $2n^2 > 2n \to n > 1$).

$$\text{Expression} < \frac{4n}{2n^2} = \frac{2}{n}$$

To ensure $\frac{2}{n} < \epsilon$, choose $N > \frac{2}{\epsilon}$.

5.3.5 C5: Chinese Remainder Theorem

System:

1. $2x \equiv 1 \pmod 5 \implies 2x \equiv 6 \implies x \equiv 3 \pmod 5$. $x = 5k+3$.
2. Substitute into second eq: $3x \equiv 2 \pmod 7$. $3(5k+3) \equiv 2 \pmod 7$ $15k+9 \equiv 2 \pmod 7$ $1k+2 \equiv 2 \pmod 7 \implies k \equiv 0 \pmod 7$.
3. Substitute back: $k = 7j$. $x = 5(7j)+3 = 35j+3$. Smallest positive solution is 3.

5.3.6 C6: Complex Geometry Interpretation

[Image of inscribed angle theorem circle]

Interpretation: The relation often describes points on the unit circle. The algebraic relation between z_1, z_2 and their arguments corresponds to the geometric property that the angle at the circumference is invariant or related to the central angle. **Formal proof:** Uses $z = e^{i\theta}$ to show the argument of the quotient corresponds to the rotation.

5.3.7 C7: Injectivity of Sets Function

Claim: f is injective iff $f(A \cap B) = f(A) \cap f(B)$. **Proof ($\Leftarrow$):** Assume the intersection property holds. Let $A = \{x\}$ and $B = \{y\}$. $f(\{x\} \cap \{y\}) = f(\emptyset) = \emptyset$. By hypothesis: $f(\{x\}) \cap f(\{y\}) = \{f(x)\} \cap \{f(y)\}$. If $f(x) = f(y)$, then $\{f(x)\} \cap \{f(y)\} = \{f(x)\} \neq \emptyset$. This contradicts the LHS being $\emptyset$ (unless $x = y$). Thus $x \neq y \implies f(x) \neq f(y)$.

5.3.8 C8: Distance Between Skew Lines

[Image of distance between skew lines projection]

1. **Normal Vector:** $\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2 = (1, 1, 0) \times (0, 1, 1)$. $\mathbf{n} = (1, -1, 1)$.
2. **Connecting Vector:** Point on L_1: $(1, 0, 0)$. Point on L_2: $(0, 1, 1)$. $\mathbf{v} = (0 - 1, 1 - 0, 1 - 0) = (-1, 1, 1)$.
3. **Projection:**

$$\text{Dist} = \frac{|\mathbf{v} \cdot \mathbf{n}|}{|\mathbf{n}|} = \frac{|(-1)(1) + (1)(-1) + (1)(1)|}{\sqrt{1^2 + (-1)^2 + 1^2}}$$

$$= \frac{|-1 - 1 + 1|}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$